Combined Science: Chemical Reactions

IGCSE Practice Paper

Total marks: 98 Estimated time: 118 minutes Questions: 33

Note: questions marked with * may stretch slightly beyond this chapter. They are included for stretch practice.

Write your answers in the boxes provided. Show working where appropriate.

Question 1. *[1 marks | Combined Science 2025 11]*

21 Which description explains why a reaction involves reduction?

A mass decreases

B oxygen is lost

C pH decreases

D temperature decreases

22 Which number determines the order of elements in the Periodic Table?

Question 2. *[1 marks | Combined Science 2025 11]*

23 Which statements about transition elements are correct?

1 They are in Group I of the Periodic Table.

2 They form coloured compounds.

3 They often act as catalysts.

4 They have low melting points.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

Question 3. *[1 marks | Combined Science 2025 13]*

A Heat liquid water to form steam.

B Mix powdered carbon and iron filings.

C Melt iron filings.

D Heat sulfur in air to make sulfur dioxide.

Question 4. *[1 marks | Combined Science 2025 22]*

and water.

What is the balanced symbol equation and correct state symbols for this reaction?

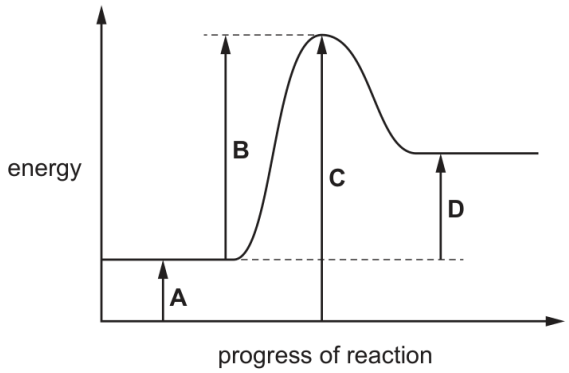
A $CaCO_3(s) + HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

B $CaCO_3(s) + HCl(l) \rightarrow CaCl_2(s) + CO_2(g) + H_2O(aq)$

C $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

D $CaCO_3(s) + 2HCl(l) \rightarrow CaCl_2(s) + CO_2(g) + H_2O(aq)$

Question 5. *[1 marks | Combined Science 2025 22]*



Question 6. *[1 marks | Combined Science 2025 12]*

20 Which statement indicates that a chemical change occurs?

A Energy is absorbed.

B There is a change in colour.

C There is a change of state.

D There is no change in mass.

Question 7. *[1 marks | Combined Science 2025 21]*

22 Which processes involve a chemical change?

1 cracking of an alkane

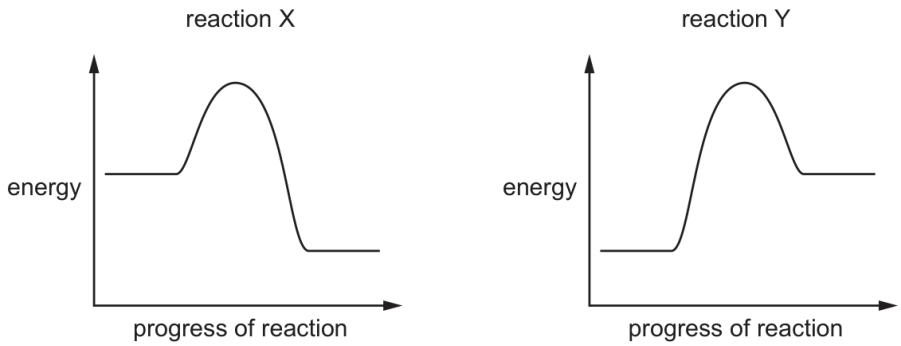
2 polymerisation of ethene

3 corrosion of iron

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Question 8. *[1 marks | Combined Science 2025 21]*

20 The reaction pathway diagrams for reaction X and for reaction Y are shown.



Which statement about the reactions is correct?

A Reaction X has a greater activation energy than reaction Y.

B Reaction X is endothermic and reaction Y is exothermic.

C The overall energy change in reaction X is much greater than in reaction Y.

D The temperature increases during reaction X and decreases during reaction Y.

*Question 9. ** *[1 marks | Combined Science 2025 21]*

A addition

B combustion

C displacement

D polymerisation

Question 10. *[1 marks | Combined Science 2025 11]*

A condensation

B evaporation

C melting

D oxidation

Question 11. *[1 marks | Combined Science 2025 11]*

The word equation for the complete combustion of pentane is shown.

pentane + oxygen → carbon dioxide + substance W

What is substance W?

A carbon monoxide

B ethane

C hydrogen

D water

Question 12. *[1 marks | Combined Science 2025 22]*

18 Which statement explains why the rate of a reaction is greater at a higher temperature?

A The activation energy is higher.

B More colliding particles have the minimum energy to react.

C The frequency of collisions between reacting particles decreases.

D There are more reacting particles per unit volume.

Question 13. *[1 marks | Combined Science 2025 22]*

24 Which process is a physical change?

A combustion of solid carbon

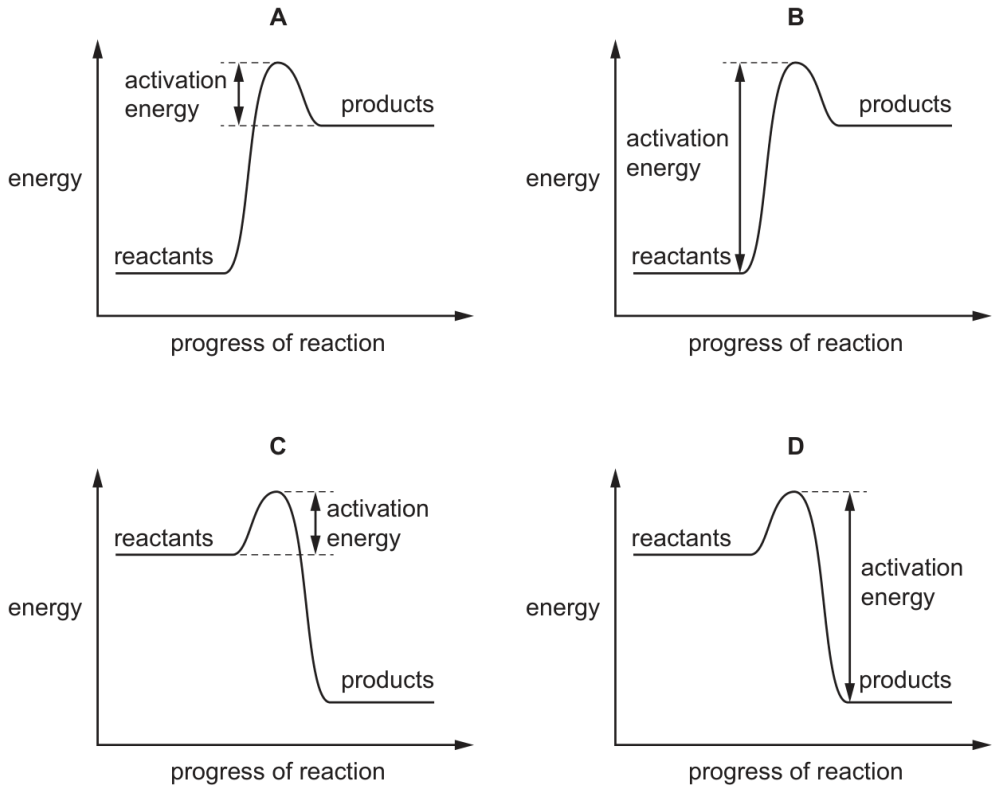
B cracking of an alkane to give an alkene

C fractional distillation of petroleum

D rusting of iron

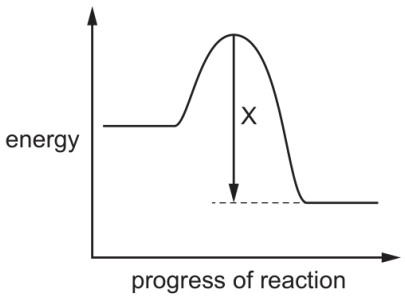
Question 14. *[1 marks | Combined Science 2025 22]*

18 Which reaction pathway diagram shows the activation energy for an exothermic reaction?



Question 15. *[1 marks | Combined Science 2025 21]*

15 A reaction pathway diagram for a reaction is shown.



Which statement describes and explains energy change X?

A Thermal energy is given out as bonds break.

B Thermal energy is given out as bonds form.

C Thermal energy is taken in as bonds break.

D Thermal energy is taken in as bonds form.

Question 16. *[1 marks | Combined Science 2025 23]*

A addition of carbon dioxide to limewater

B addition of sodium to water

C electrolysis of molten lead(II) bromide

D separation of petroleum by fractional distillation

Question 17. *[1 marks | Combined Science 2025 23]*

Which change decreases the rate of this reaction?

A adding a catalyst to the reaction mixture

B increasing the concentration of the hydrochloric acid

C increasing the size of the zinc lumps

D increasing the temperature of the hydrochloric acid

18 The equation for the complete combustion of ethane is shown.

Question 18. *[1 marks | Combined Science 2025 23]*

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$$

Which statement about this reaction is correct?

A Ethane is oxidised because it gains oxygen.

B Ethane is reduced because carbon–hydrogen bonds are broken.

C It is endothermic because thermal energy is transferred to the surroundings.

D It is exothermic because thermal energy is taken in from the surroundings.

Question 19. * [1 marks | Combined Science 2025 21]

19 Which metal reacts most quickly with cold water?

A calcium

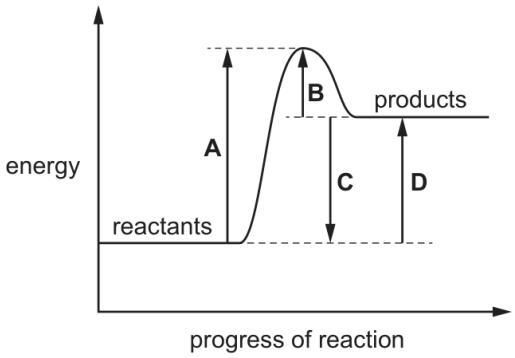
B magnesium

C potassium

D sodium

Question 20. [1 marks | Combined Science 2025 23]

19 Which arrow on the reaction pathway diagram represents the overall energy change for the reaction?



Question 21. [1 marks | Combined Science 2025 12]

21 Which two substances form when magnesium reacts with steam?

A magnesium hydroxide and hydrogen

B magnesium hydroxide and oxygen

C magnesium oxide and hydrogen

D magnesium oxide and water

22 Anhydrous copper(II) sulfate is used to test for water.

Question 22. [1 marks | Combined Science 2025 23]

23 The equations describe the reactions that happen when iron is extracted in the blast furnace.

$$C + O_2 \rightarrow CO_2$$

$$C + CO_2 \rightarrow 2CO$$

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

What is a description of one of these reactions?

A Carbon dioxide is oxidised to form carbon monoxide.

B Iron(III) oxide is reduced by carbon dioxide.

C Carbon reacts with carbon monoxide to form carbon dioxide.

D Carbon monoxide removes oxygen from iron(III) oxide.

24 The structures of some compounds are shown.

Question 23. *[3 marks | Combined Science 2025 31]*

6 **(a)** Fig. 6.1 shows the combustion of magnesium ribbon in air.

A white powder of magnesium oxide is formed.

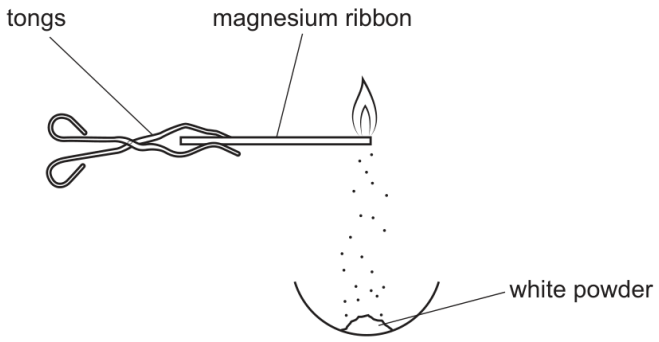


Fig. 6.1

(i) Name the gas that reacts with the magnesium.

..... [1]

(ii) Magnesium oxide is a metallic oxide.

Circle the term that describes this type of oxide.

acidic **aqueous** **basic** **carbonate** **solvent**

[1]

(iii) This reaction is exothermic.

Describe what is meant by exothermic.

.....

..... [1]

Question 24. *[3 marks | Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

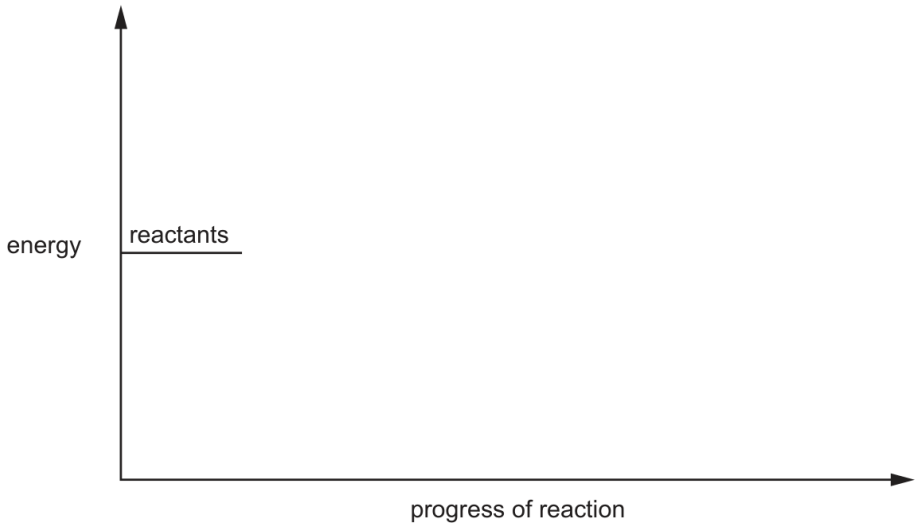


Fig. 5.1

[3]

[Total: 9]

Question 25. *[5 marks | Combined Science 2025 42]*

5 **(a)** Calcium oxide, CaO, is used to make cement.

Calcium oxide is formed by the decomposition of calcium carbonate, $CaCO_3$, at a very high temperature.

The equation for the reaction is shown.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

(i) Suggest why making cement contributes to global warming.

.....

.....

.....

..... [2]

(ii) The reaction is endothermic.

Explain what is meant by an endothermic reaction.

Use ideas about making bonds and breaking bonds in your answer.

.....

.....

.....

.....

..... [3]

Question 26. *[6 marks | Combined Science 2025 32]*

6 Hydrogen peroxide, H_2O_2, decomposes into water and oxygen when a catalyst is added.

(a) Balance the symbol equation for the reaction.

$$\ldots\ldots\ldots\ldots H_2O_2 \rightarrow \ldots\ldots\ldots\ldots H_2O + O_2$$

[

(b) The oxygen is collected and tested.

Describe the test for oxygen gas.

State the observation for the positive result.

test .....

observation .....

[

(c) Anhydrous cobalt(II) chloride is used to test for water.

State the colour observed for a positive result.

..... [

(d) Describe what is meant by a catalyst.

.....

.....

.....

..... [

(e) The temperature of the hydrogen peroxide is increased. All other conditions stay the same.

Predict the effect of this change on the reaction.

..... [

[Total:

Question 27. *[6 marks | Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

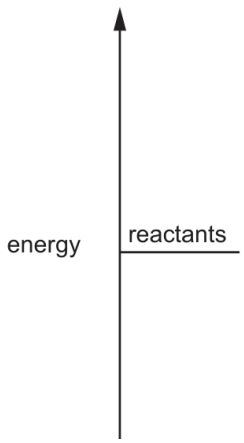
A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.



5 The reaction between aluminium and iron(III) oxide is used to join iron railway lines together.

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(l) + Al_2O_3(s)$$

Molten iron is produced in the reaction. Molten iron becomes solid when it cools.

(a) **(i)** State how the equation shows that **molten** iron is produced.

..... [1]

(ii) Use the equation to explain why this reaction is a redox reaction.

.....

.....

..... [2]

(iii) Complete the ionic equation for this reaction.

$Al + Fe^{3+} \rightarrow$ ..... + ... [2]

(b) Explain the meaning of the symbol (III) in iron(III) oxide.

.....

..... [1]

Question 28. *[6 marks | Combined Science 2025 33]*

Use words from the list.

Each word may be used once, more than once or not at all.

anions **cations** **covalent** **electrons** **ionic**

neutrons **neutralisation** **oxidation** **protons** **reduction**

During the reaction, magnesium atoms lose ..... to form positive ions.

These positive ions are known as ..... .

Oxygen atoms from the water molecules form negative ions.

Positive ions and negative ions attract each other and form ..... bonds.

During the reaction, magnesium atoms gain oxygen. This type of reaction is ..... .

[4]

(b) The student repeats the experiment using copper instead of magnesium.

Predict what the student observes during the reaction.

Explain your answer.

observation .....

.....

explanation .....

.....

[2]

[Total: 9]

Question 29. *[7 marks | Combined Science 2025 41]*

(c) Fig. 4.2 is a reaction pathway diagram for a reaction catalysed by copper.

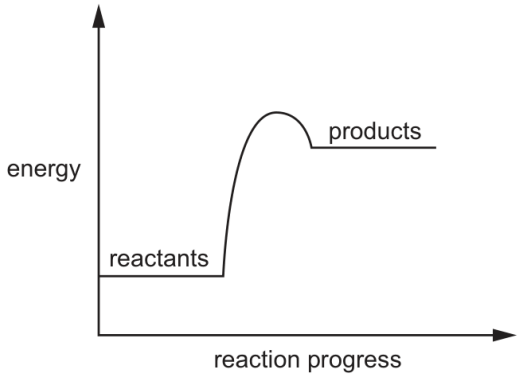


Fig. 4.2

(i) Identify the type of chemical reaction shown in Fig. 4.2.

Explain your answer.

type of chemical reaction .....

explanation .....

.....

..... [2]

(ii) Draw an arrow on Fig. 4.2 to show the activation energy for this reaction. [1]

(iii) Define activation energy.

.....

.....

.....

..... [2]

(iv) Describe what is meant by a catalyst.

.....

.....

.....

..... [2]

[Total: 11]

Question 30. *[9 marks | Combined Science 2025 43]*

4 A student investigates the rate of reaction between large pieces of magnesium carbonate, MgCC and dilute hydrochloric acid, HC*l*, using the apparatus shown in Fig. 4.1.

The student records the time taken to produce 10 cm^3 of gas.

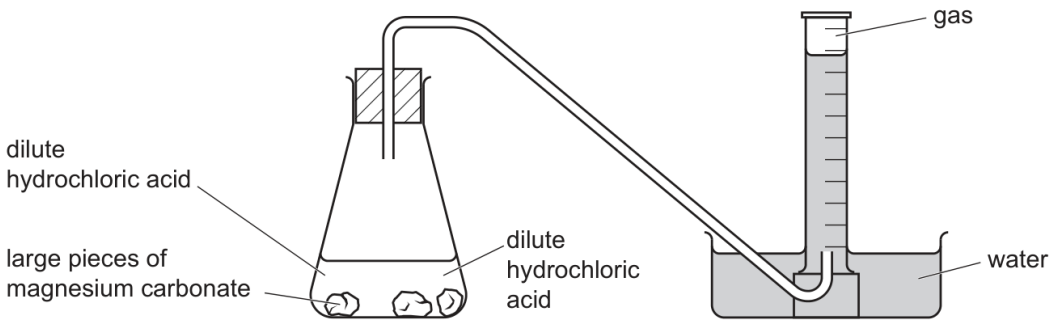


Fig. 4.1

(a) Three products are formed in the reaction: magnesium chloride $MgCl_2$, a gas and one oth product.

Give the balanced symbol equation for this reaction.

..... [

(b) The student repeats the experiment using:

- small pieces of magnesium carbonate
- magnesium carbonate powder.

All other conditions are kept constant.

Table 4.1 shows the time taken to produce 10 cm^3 of gas and the rate of reaction.

Table 4.1

magnesium carbonate	**time taken to produce 10 cm^3 of gas in s**	**rate of reaction in cm^3 per s**
large pieces	15.2	0.66
small pieces	8.3	1.2
powder	5.1	

(i) Calculate the rate of reaction in cm^3 per s for magnesium carbonate powder.

Write your answer in Table 4.1. [

Question 31. *[9 marks | Combined Science 2025 32]*

4 A student investigates the reaction between magnesium and steam.

Fig. 4.1 is a diagram of the apparatus used in the investigation.

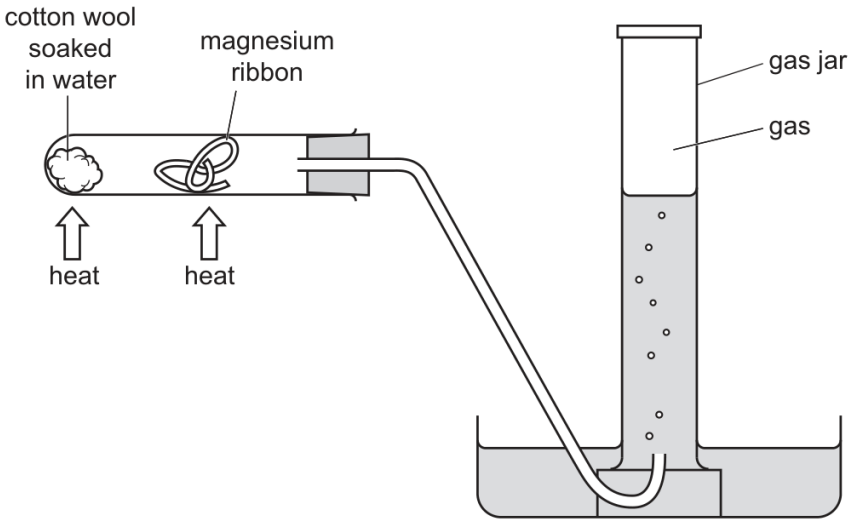


Fig. 4.1

(a) The reaction produces magnesium oxide and a gas.

(i) Complete the word equation for this reaction.

	+		⟶	magnesium oxide	+	

[

(ii) The gas jar in Fig. 4.1 collects the gas, but it does **not** allow the volume to b measured directly.

Name **one** piece of apparatus that allows the volume of gas to be measured directly.

..... [

Question 32. *[9 marks | Combined Science 2025 33]*

6 Scientists are investigating the use of iron as a fuel in cars.

Iron reacts with oxygen, as shown in equation 1. This reaction releases energy.

The iron oxide formed is converted back to iron, as shown in equation 2.

equation 1 .....$Fe(s)$ + ...O_2(.....) $\rightarrow$ $2Fe_2O_3(s)$

equation 2 $Fe_2O_3(s) + 3H_2(g) \rightarrow 2Fe(s) + 3H_2O(l)$

(a) Balance equation 1 and add the missing state symbol for oxygen. [

(b) State the name given to a chemical reaction that releases thermal energy.

..... [

(c) Use the substances in equation 1 and equation 2 to answer the following questions.

(i) Identify the compound that exists as simple molecules.

..... [

(ii) Identify the transition element.

..... [

(d) Explain why equation 1 shows that iron is oxidised.

.....

..... [

(e) Most cars burn fuels that contain carbon.

Explain why using iron as a fuel may cause less harm to the environment than using fue that contain carbon.

.....

.....

.....

..... [

[Total:

Question 33. *[13 marks | Combined Science 2025 62]*

2 A student investigates the reaction between aluminium and aqueous copper(II) chloride.

Aqueous copper(II) chloride is a blue solution.

Aluminium is a grey solid.

(a) Procedure

The student:

- assembles the apparatus shown in Fig. 2.1

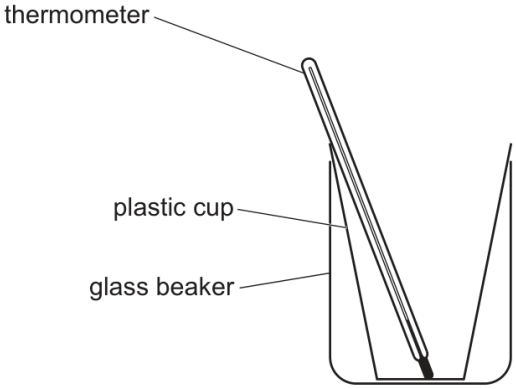


Fig. 2.1

- pours 25 cm^3 of aqueous copper(II) chloride into the plastic cup
- records in Table 2.1, the initial temperature of the aqueous copper(II) chloride
- adds 0.5 g of aluminium to the aqueous copper(II) chloride
- stirs the mixture in the plastic cup
- records in Table 2.1, the highest temperature reached by the reaction mixture.

(i) Fig. 2.2 shows the temperature readings for this experiment.

Fig. 2.2

Record in Table 2.1, these temperatures to the nearest 0.5 °C.

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